THE UNIVERSITY OF HONG KONG

BACHELOR OF ENGINEERING

DEPARTMENT OF CIVIL ENGINEERING

CIVL1113: ENGINEERING MECHANICS AND MATERIALS

May 8th, 2024 9:30 AM – 12:30 PM (3 hours)

Answer any **5 (FIVE)** questions from Section A and B.
All questions carry equal marks.
Marks for sub-questions (if any) are given.

Use a separate answer book for each section.

Use of Calculators:

Only approved calculators as announced by the Examinations Secretary can be used in this examination. It is candidates' responsibility to ensure that their calculator operates satisfactorily, and candidates must record the name and type of the calculator used on the front page of the examination script.

SECTION A

Q1(a) A horizontal beam ABC is subject to uniformly distributed load 10 kN/m and an external moment of 10kN×m as shown in Figure 1(a).

(i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

(ii) Draw the bending moment diagram (BDM) with pertinent values. Noe that bending moment should be drawn on the **tensile** side. **(5 marks)**

(b) A frame is subject to a horizontal distributed load in the column AB and a vertical force applied at the beam BC, as illustrated in Figure 1(b). A is fixed, B is pin connection, C is roller. The vertical force of 900N is located at 1.2 m from B.

(i) Draw the shear force diagram (SFD) with the sign labels and pertinent values. Note that positive shear is along the **clockwise** direction. **(5 marks)**

(ii) Draw the bending moment diagram (BDM) with pertinent values. Noe that bending moment should be drawn on the **tensile** side. **(5 marks)**

Q2 (a) A cantilever beam has section as shown in Figure 2(a) and Figure 2(b). The density of beam is 7850kg/m3. Gravity acceleration is 9.8m/s2. The beam height is 1864 mm, flange width is 450 mm, flange thickness is 32mm, web thickness is 2 mm. Only consider self-weight of the beam. The span length of the beam is 40 m. The allowable stress is 80 MPa. Check the safety of the beam based on von Mises criterion. Neglect buckling failure in this problem. **(10 marks)**

(b) The point A is subject to horizontal tensile stress of 40 MPa, vertical compressive stress of 160 MPa and the magnitude of shear stress is 54 MPa as shown in Figure 2(c).

(i) Obtain **principal** tensile stress and **principal** compressive stress at the point. Draw the direction where the principal compressive stress will happen. **(5 marks)**

(ii) Check the strength condition according to the von Mises failure criteria. The allowable von Mises stress is $[\sigma]$=180 MPa. **(5 marks)**

Q3 (a) A shaft is subject to a torque of T=20kN×m as shown in Figure 3(a). The shaft is made of solid round section with diameter d. The elastic modulus is 200 GPa and Poisson ratio is 0.3. The allowable shear stress is 48 MPa and the allowable rate of twist is 0.001 rad/m. What is the minimum required diameter d if the shaft needed to meet all these requirements. **(10 marks)**

(b) Consider the column problem as shown in Figure 3(b). A rectangular column with cross-sectional dimensions b and h is pin supported at ends A and C (see figure). Pin at A and C are free to rotate in all directions. At mid-height section, the column is restrained in the plane of the figure, but is free to deflect perpendicularly to the plane of the figure. The total column of truss is L=30 m. The elastic modulus is 200 GPa and Poisson ratio is 0.3. The section height is h=200mm, the section width is b=120 mm. Consider both in-plane and out-of-plane buckling and use the minimum load P between these two failure modes.

Check the load P that will cause buckling in the member based on Euler's buckling theory. Calculate the compressive stress of the column at the buckling load. Assume safety factor is 1.0. **(10 marks)**

Hint for 3(a) and 3(b): The polar moment of inertia of solid round section is $\frac{\pi d^4}{32}$, where d is the diameter of cross-section. The moment of inertia of solid round section with respect to the neutral axis is $\frac{\pi d^4}{64}$, where d is the diameter of the round section. According to Euler's formula, the critical load for a slender column with span L is

$$P_{cr} = \frac{\pi^2 EI}{(\mu l)^2} \quad \text{(Euler Formula)}$$

The coefficient μ is 1.0 for both pinned ends, μ is 2.0 for one fixed end and one free end, μ is 0.7 for one fixed end and one pinned end, and μ is 0.5 for both fixed ends.

SECTION B

Q4 (a) Both timber and plywood were commonly used as formwork for concrete casting. However, nowadays steel formwork is more popular. Why? **(8 marks)**

(b) Sketch the various layers forming a flexible pavement and write an account of the bituminous mixtures for the various layers. **(6 marks)**

(c) Discuss the fire resistance of concrete and structural steel. Which one is more fire resistant? Any measures to improve fire resistance of the weaker one? **(6 marks)**

Q5 (a) Explain the consequences and precautions for hot weather concreting. **(8 marks)**

(b) Discuss the porosity in concrete and the biggest factor affecting the porosity of concrete. **(6 marks)**

(c) Explain characteristics of good quality aggregates. **(6 marks)**

Q6 (a) Explain what the three common crystal defects in metals are. Draw a sketch of each of the three common crystal defects. **(8 marks)**

(b) Explain the corrosion protection methods of structural steel. **(6 marks)**

(c) Other than strength, which properties of engineering materials are worth consideration? Explain briefly with examples. **(6 marks)**

FIGURES

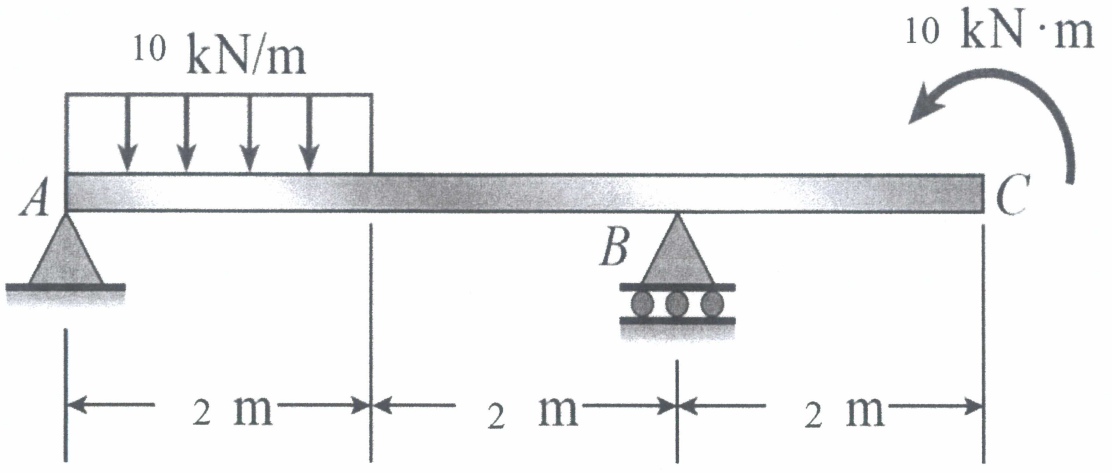


Figure 1(a). Beam

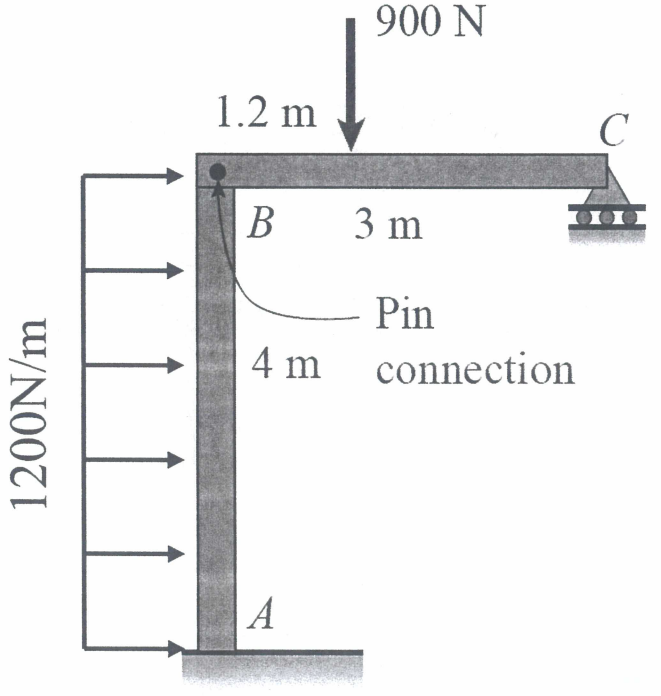


Figure 1(b). Frame

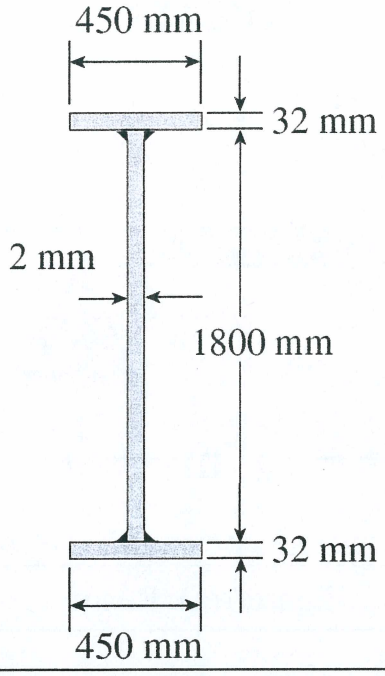


Figure 2(a). Cross-section of beam

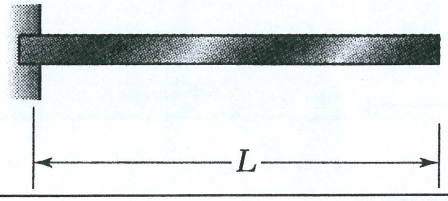


Figure 2(b). Beam under self-weight only

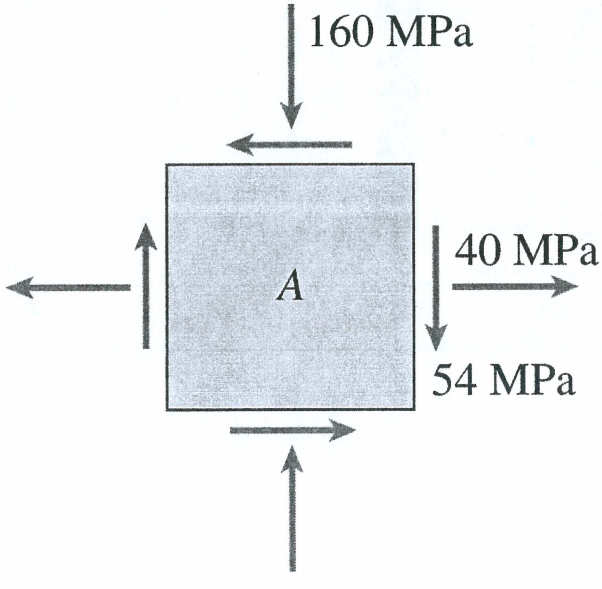


Figure 2(c). Stress element

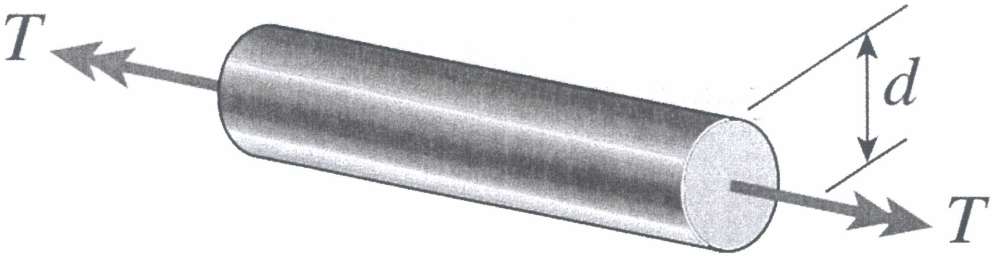


Figure 3(a). Shaft under pure torsion

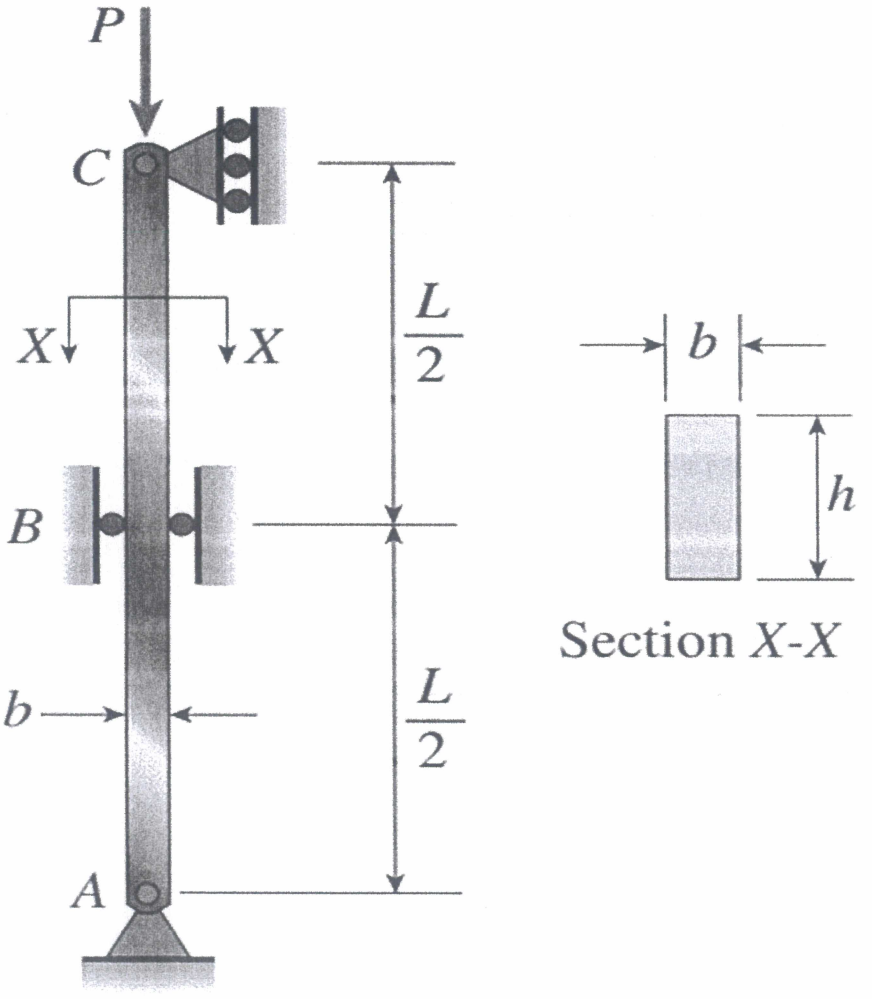


Figure 3(b). Column

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